SUBSTITUTE SPECIFICATION

METHOD AND SYSTEM FOR DIRECT ANALOG SIGNAL ENCODING ON A HIGH-CARDINALITY PARALLEL INTERFACE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of and priority to the portfolio of inventions described in the publication entitled "The Architecture of Control," ISBN 978-1-97-148129-6, the contents of which are hereby incorporated by reference in their entirety. The method described herein is an improvement upon the high-cardinality parallel data bus described therein.

FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

Not applicable.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates generally to digital data communication, signal processing, and high-bandwidth parallel interface architectures, and more specifically to methods and systems for directly encoding analog signal waveforms into multi-state, high-cardinality parallel interfaces without intermediate binary quantization bottlenecks.

2. Description of the Related Art

The representation of analog signals, such as audio, video, sensor data, and continuous physical waveforms, in digital systems has traditionally relied on a multi-step conversion architecture. Conventionally, an analog-to-digital converter (ADC) samples the analog waveform at a specific sampling frequency (e.g., Nyquist rate). Each sample's instantaneous amplitude is quantized to the nearest value within a discrete set of linear or logarithmic binary levels (e.g., 16-bit or 24-bit

PCM). This quantized value is subsequently serialized and encoded as a multi-bit binary word. The resulting sequence of binary numbers is stored in a structured file container (such as WAV, FLAC, or MP4) or transmitted serially over electronic interconnects.

While foundational to modern computing, this conventional digitization architecture introduces substantial structural inefficiencies. The mandatory conversion of continuous physical phenomena into serial, binary bitstreams imposes severe serialization bottlenecks, incurs latency, requires high-frequency clocking, and fundamentally fails to utilize the intrinsic parallel bandwidth capabilities of high-radix and high-cardinality transmission systems, particularly optical and multi-wavelength interfaces. A need therefore exists for an architectural framework capable of directly mapping analog physical waveforms into parallel high-cardinality state spaces.

BRIEF SUMMARY OF THE INVENTION

The present invention overcomes the limitations of the prior art by providing a method and system for the direct encoding of analog signals onto a high-cardinality, parallel interface. Rather than serializing and quantizing samples into binary numerical words, the system samples an analog signal and maps discrete waveform characteristics directly to the discrete physical states of a high-cardinality state space supported by a parallel physical bus.

In a preferred optical embodiment, the instantaneous amplitude or frequency characteristic of an analog sample is mapped directly to a specific wavelength, phase, or spatial state of light within an optical bus. A sequence of such samples is converted into a structured array of multi-state optical elements, forming a frame of high-cardinality data. This frame can be transmitted in a single parallel pulse or transfer cycle across an optical interface, or stored within an optical medium. Bypassing intermediate binary conversion unlocks unprecedented density and throughput, enabling complex analog media to be transmitted and stored in compact high-radix frames.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a conceptual block diagram contrasting traditional analog-to-digital conversion with direct high-cardinality analog signal encoding according to the present invention.

FIG. 2 is a schematic diagram illustrating the direct mapping of an analog waveform to a sequence

of high-cardinality data states according to the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

The present invention provides a direct-to-state encoding architecture that eliminates conventional intermediate binary analog-to-digital conversion stages.

Referring to FIG. 1, the fundamental architectural contrast between conventional signal processing and the present invention is shown. In conventional architectures (FIG. 1, Top), an incoming continuous analog signal (101) is input to an analog-to-digital converter (102). The ADC quantizes each sample into a multi-bit binary value (103), which is subsequently serialized into an electronic bitstream (104) and transmitted across a serial bus (105).

In contrast, under the system of the present invention (FIG. 1, Bottom), the analog signal (101) is sampled at discrete time intervals by a sampling module (106). The sampled values are delivered directly to a direct encoding module (107). The direct encoding module (107) bypasses binary serialization and maps the continuous sample characteristic (such as voltage amplitude, frequency, or phase) directly into a discrete multi-state symbol chosen from a high-cardinality state space (108). The resulting high-cardinality state is output directly to a parallel interface (109), such as an optical multi-wavelength bus, spatial light modulator, or multi-core optical waveguide.

Referring to FIG. 2, the direct mapping operation is illustrated in detail. An analog waveform (201) is sampled at discrete temporal intervals (T_1, T_2, T_3, T_4). The instantaneous amplitude of the waveform at each interval (A_1, A_2, A_3, A_4) is determined. A direct mapping function (202) assigns each amplitude value to a corresponding unique state within a high-cardinality physical state space.

In an exemplary optical embodiment, each state is defined by a specific optical wavelength or color (for example, state S_1 maps to a first discrete wavelength, state S_2 maps to a second discrete wavelength, etc.). The resulting output is a spatial or temporal sequence of high-cardinality physical states (203) that directly and losslessly represent the sampled analog signal. This sequence of states can be transmitted in parallel across an optical interconnect as a single coherent data frame, achieving extreme bandwidth efficiency without the computational and thermal overhead of binary

serialization.

INVENTOR STATEMENT UNDER 37 CFR § 1.125(b)

Pursuant to 37 CFR § 1.125(b), the undersigned Applicant hereby certifies that this substitute specification includes no new matter.

Respectfully submitted,

/Reinaldo Javier Aguiar Marcano/
Reinaldo Javier Aguiar Marcano
Inventor and Pro Se Applicant
Mailing Address: 2302 Britton Ridge Drive, Katy, Texas 77494
Telephone: (202) 599-3979
Email: `rj.aguiar@gmail.com` (USPTO Account)
Email: `rj.aguiar.m@gmail.com`
Email: `reinaldo.aguiar.marcano@isbn9781971481296.com`
Date: October 10, 2026